

Macroeconomic Determinants of Box Office Revenue

A time-series analysis of consumer spending, monetary policy, and theatrical demand, 2000–2025

PRIMARY
DRIVER

PCE

METHOD
STACK

**VAR ·
VECM**

COINTEGRATION

1 vector

SAMPLE

312 mo.

Prepared by Tanmay Srivastava · M.S. Applied Economics (Entertainment Economics), University of Cincinnati. Originally submitted as ECON 8011 coursework under Dr. Olivier Parent and reformatted as an analytical work sample. Series are monthly observations of domestic box office revenue, personal consumption expenditures (PCE), and the effective federal funds rate. Modeling performed in R (*vars*, *urca*, *tsDyn*). Figures reconstructed for presentation; underlying estimates unchanged.

Theatrical demand rides the macro cycle—but corrects on its own terms

What moves box office revenue, over what horizon, and where the adjustment actually happens.

This analysis tests whether macroeconomic conditions materially shape box office performance, using a monthly panel spanning 2000–2025 and a layered time-series framework—dynamic regression, a three-variable vector autoregression (VAR), and a vector error correction model (VECM). The headline result is not simply that consumption matters; it is **where the system absorbs disequilibrium**. When box office revenue and consumer spending drift from their long-run relationship, it is the broad economy—not the film industry—that adjusts. Theatrical demand tracks the macro cycle in the short run while retaining strong, idiosyncratic internal dynamics driven by release slates, seasonality, and shocks.

PCE → BOX OFFICE

Strong

FED FUNDS → BOX OFFICE

Weak

ERROR CORRECTION IN BOX OFFICE EQN.

None

OWN-SHOCK SHARE OF BOX OFFICE VAR.

~87%

Three findings

- 1 Consumer spending is the dominant, robust driver.** Growth in PCE predicts box office growth in both the dynamic regression and the VAR; Granger causality runs strongly from PCE to revenue. A positive PCE shock lifts revenue immediately, then decays—transient but economically meaningful.
- 2 Interest rates matter, but faintly.** The federal funds rate enters with a small, statistically detectable effect and limited predictive power—consistent with monetary policy acting on theatrical demand only indirectly, through discretionary income.
- 3 The economy absorbs the imbalance, not the box office.** The Johansen test finds one cointegrating vector, but the error-correction term is significant only in the PCE equation. Box office revenue does not re-equilibrate toward the long-run relationship—its volatility is industry-specific.

TAKEAWAY

Box office is macro-sensitive in the short run and macro-independent in its correction dynamics. For forecasting and greenlight/financing decisions, consumer-spending conditions are a leading input—but the industry's own release cycles dominate the variance and cannot be hedged with macro variables alone.

A volatile dependent variable against two smoother macro series

Monthly observations, 2000–2025, with a decisive structural break in 2020.

The dataset comprises monthly domestic box office revenue (BO_t), personal consumption expenditures (PCE_t), and the effective federal funds rate (FFR_t). The three series behave very differently: box office is highly volatile with strong seasonality; PCE follows a smooth upward trend; and the funds rate moves in slow policy cycles. COVID-19 in 2020 is a sharp structural break—revenue collapses toward zero while PCE dips only briefly.

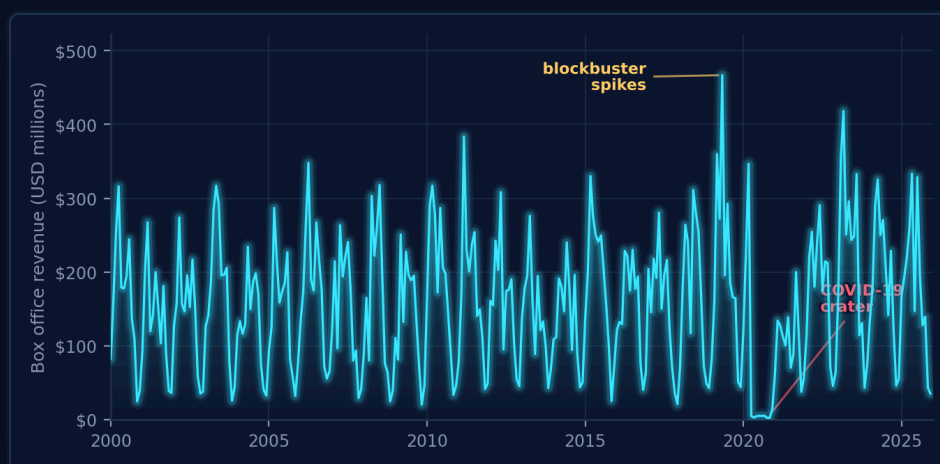


Figure A1. Domestic box office revenue, monthly. Pronounced seasonal spikes and the 2020 collapse toward zero.

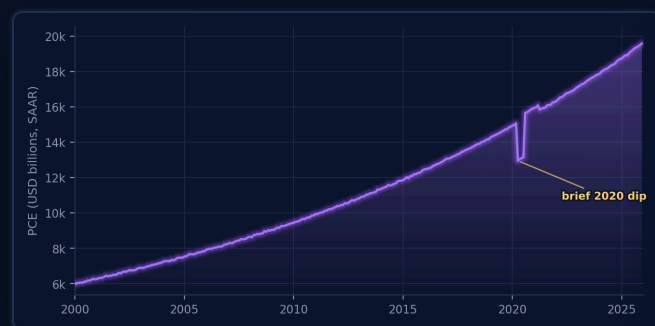


Figure A2. PCE—smooth secular uptrend, shallow 2020 dip.



Figure A3. Effective federal funds rate—cyclical policy regimes.

Because all three series trend, the analysis works in log-differences for revenue and PCE and first differences for the rate—yielding growth-rate interpretations and approximate stationarity: $\Delta \log(BO_t)$, $\Delta \log(PCE_t)$, ΔFFR_t .

TAKEAWAY

The transformed series shed their trends and are suitable for VAR/VECM estimation; the 2020 break is the primary source of residual diagnostic strain noted later.

Levels persist; differences decay quickly

Autocorrelation structure justifies the differencing strategy and lag selection.

Autocorrelation (ACF) and partial autocorrelation (PACF) functions confirm the transformation choice. In levels, log box office shows slow, persistent decay—the signature of a trending, near-integrated series. After differencing, autocorrelation collapses rapidly toward the confidence bands, indicating stationarity and modest residual seasonal structure. The same pattern holds for PCE and the funds rate.

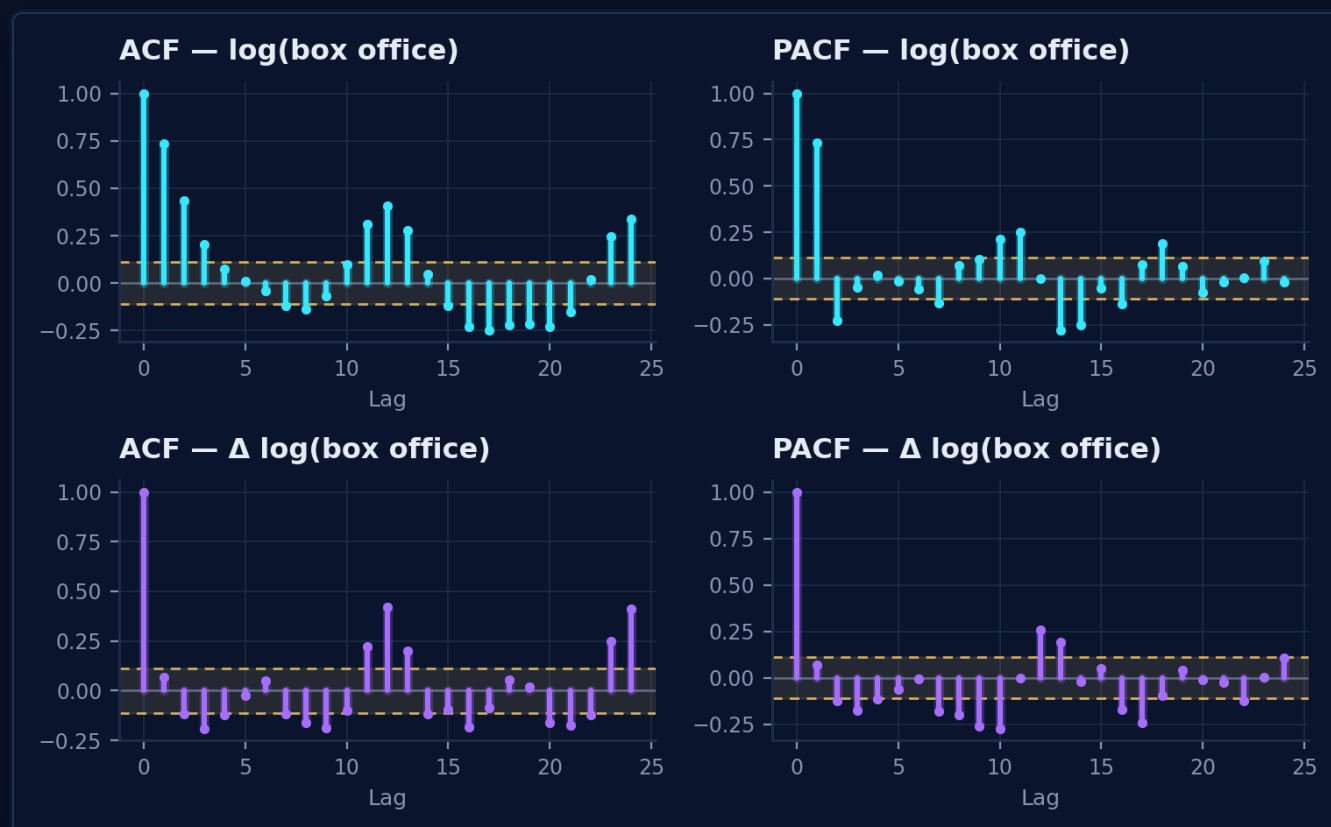


Figure C1–C2. ACF and PACF of log box office (top) and its first difference (bottom). Slow decay in levels versus rapid decay after differencing. Dashed bands are approximate 95% limits.

These lag structures inform a VAR lag length of three, balancing fit against parsimony, and are consistent with the predictive relationships formalized by the Granger tests in Section 4.

TAKEAWAY

Differencing is well-justified: it removes persistence without stripping the short-run dynamics the models are meant to capture.

From dynamic regression to a full error-correction system

Four nested tools, each relaxing an assumption of the last.

Dynamic regression

The baseline models box office growth as a function of its own lags, lagged PCE growth, lagged rate changes, and monthly seasonal dummies D_m —essential given predictable summer and holiday peaks. The coefficients carry clean economic interpretations.

$$\Delta \log(BO_t) = \alpha + \sum_i \beta_i \Delta \log(BO_{t-i}) + \sum_i \gamma_i \Delta \log(PCE_{t-i}) + \sum_i \delta_i \Delta FFR_{t-i} + \sum_m \theta_m D_m + \varepsilon_t$$

γ_i measure sensitivity to consumer spending; β_i capture momentum / mean-reversion; δ_i capture monetary conditions.

Vector autoregression (VAR)

The regression treats macro variables as one-way predictors. To allow feedback—box office may itself reflect broad consumption patterns—all three variables are treated as endogenous in a three-lag VAR on the stationary vector $y_t = [\Delta \log BO, \Delta \log PCE, \Delta FFR]'$. Lag length is chosen by information criteria, balancing fit against parsimony.

$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + A_3 y_{t-3} + u_t$$

WHY THIS LADDER

Each step tests a sharper question: the regression asks whether macro conditions predict revenue; the VAR asks whether the relationship runs both ways; cointegration and the VECM (next page) ask whether a long-run equilibrium exists and *who* adjusts to restore it.

Testing precedence, equilibrium, and adjustment

Granger causality, the Johansen test, and the error-correction system.

Granger causality

Granger tests assess predictive precedence: if the lagged terms of one variable are jointly significant in the equation for another, the first is said to Granger-cause the second. This is predictive usefulness, not structural causation—an important caveat carried through the results.

Cointegration

Differencing removes long-run relationships along with trends. If the level series share a common stochastic trend, they may be cointegrated—implying a stable equilibrium. The Johansen procedure tests for this; the analysis finds at least one cointegrating vector, consistent with the shared trending behavior visible in the level plots.

$$\phi_1 \log(BO_t) + \phi_2 \log(PCE_t) + \phi_3 FFR_t + c = 0$$

Vector error correction (VECM)

Given cointegration, the VECM nests both long-run equilibrium and short-run dynamics. The error-correction term $\beta'y_{t-1}$ measures the deviation from equilibrium; the loading vector α measures how quickly each variable adjusts to close that gap.

$$\Delta y_t = \alpha \beta' y_{t-1} + \sum_i \Gamma_i \Delta y_{t-i} + u_t$$

The sign and significance of α by equation reveal which variable does the adjusting—the study's central result, reported in Section 06.

TAKEAWAY

The loading vector α is the crux: a significant loading in one equation and not another tells us whether the film industry or the broader economy bears the burden of restoring long-run balance.

Consumption predicts revenue; rates barely register

Dynamic regression, VAR, and Granger causality point the same direction.

The dynamic regression shows a strong positive association between PCE growth and box office growth: higher aggregate consumption translates into higher movie-going demand. Lagged box office terms are negative and significant—evidence of mean reversion, as unusually high or low months correct over release cycles. Rate changes are included but statistically weak in the short run, reflecting indirect monetary transmission.

RELATIONSHIP	DIRECTION	STRENGTH	INTERPRETATION
PCE growth → box office growth	Positive	STRONG	Primary macro channel
Lagged box office (own)	Negative	MODERATE	Mean reversion / release cycles
Fed funds change → box office	Weak –	FAINT	Indirect, via discretionary income
Box office → PCE (feedback)	Positive	SOME	Not strictly one-way

Granger causality corroborates the regression: changes in PCE substantially predict box office income, with some evidence of feedback from box office to PCE, and lower predictive power for the funds rate. The relationship between the macro economy and theatrical demand is strong but not purely unidirectional.

TAKEAWAY

Across three independent tests, consumer spending is the consistent predictor of box office growth. Monetary policy's role is real but second-order.

A consumption shock lifts revenue immediately, then fades

Tracing a one-standard-deviation shock through the VAR.

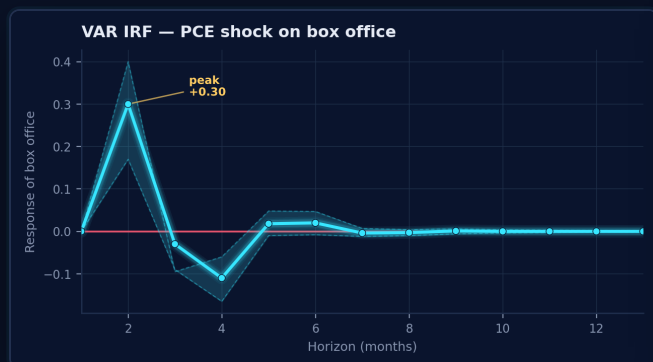


Figure D1. Response of box office to a PCE shock: a sharp positive jump that decays to baseline within roughly six months.

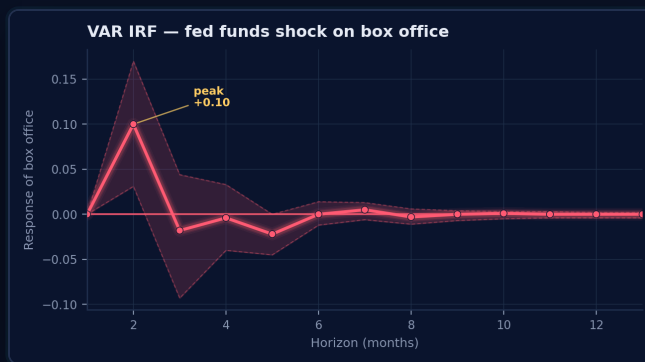


Figure D2. Response of box office to a funds-rate shock: smaller, shorter-lived, and quickly indistinguishable from zero.

The contrast is the point. A positive PCE shock produces an immediate, economically large increase in box office revenue that dissipates over the following months—transient but significant. A funds-rate shock generates a muted, less persistent response, reinforcing the finding that monetary policy reaches theatrical demand only indirectly. Bootstrap confidence bands (shaded) confirm the PCE response is cleanly separated from zero at the peak, while the rate response is not.

TAKEAWAY

For near-term revenue forecasting, current consumption conditions carry real signal; rate moves are largely noise at the industry level.

Strong internal dynamics, meaningful macro overlay

Where box office variance comes from—and who closes the equilibrium gap.

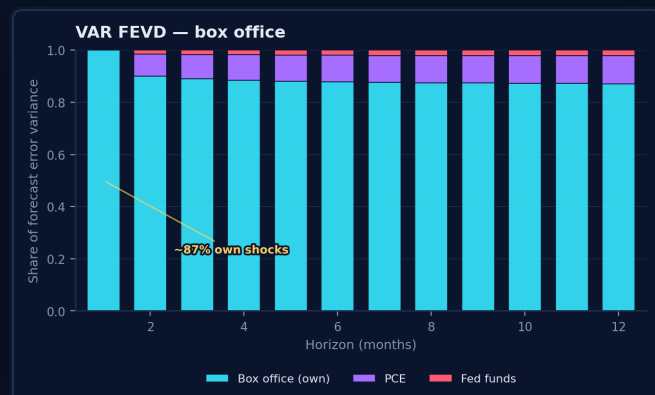


Figure D3. Forecast error variance decomposition for box office. Own shocks dominate at roughly 87%, with PCE contributing a non-trivial and growing share.

The variance decomposition shows box office revenue is explained mostly by its own shocks—confirming strong industry-specific dynamics—while PCE accounts for a modest, rising fraction over the forecast horizon. The film business has powerful internal momentum, yet remains meaningfully exposed to the aggregate economy.

The VECM delivers the study's sharpest result. The error-correction term is significant in the **PCE equation** but **absent from the box office equation**. When the system deviates from its long-run equilibrium, consumer spending adjusts to restore it—box office revenue does not. The imbalance is absorbed by the broad economy rather than corrected by the film industry, consistent with PCE representing underlying fundamentals while box office is the more erratic, industry-driven series.

TAKEAWAY

Box office is a taker, not an adjuster, in the long-run relationship. It responds to macro conditions but does not self-correct toward them—an important distinction for anyone modeling theatrical revenue over multi-year windows.

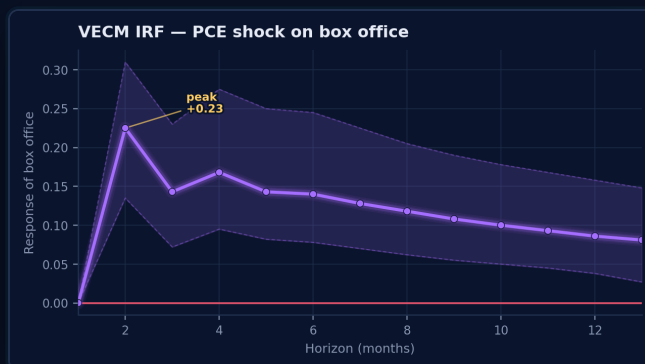


Figure E1. VECM impulse response: under the error-correction system, a PCE shock produces a larger and more persistent box office response than the differenced VAR.

Short-horizon forecasts and honest limitations

Out-of-sample projection and where the model strains.

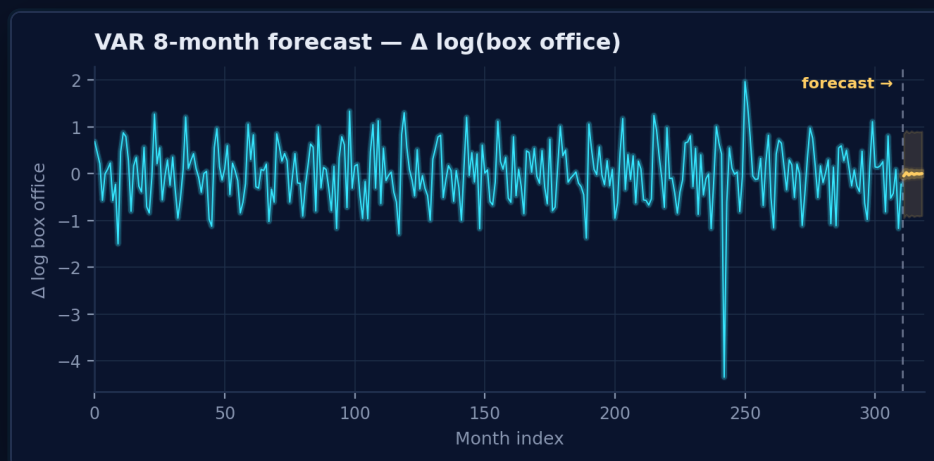


Figure F1. Eight-month-ahead VAR forecast of box office growth with bootstrap prediction band. The forecast reverts toward the seasonal mean; wide bands reflect the series' intrinsic volatility.

Both VAR and VECM produce short-horizon forecasts that revert toward the seasonal mean, with prediction intervals wide enough to reflect the genuine volatility of theatrical revenue—informative for the direction and magnitude of near-term movements, not for pinpointing individual blockbuster months.

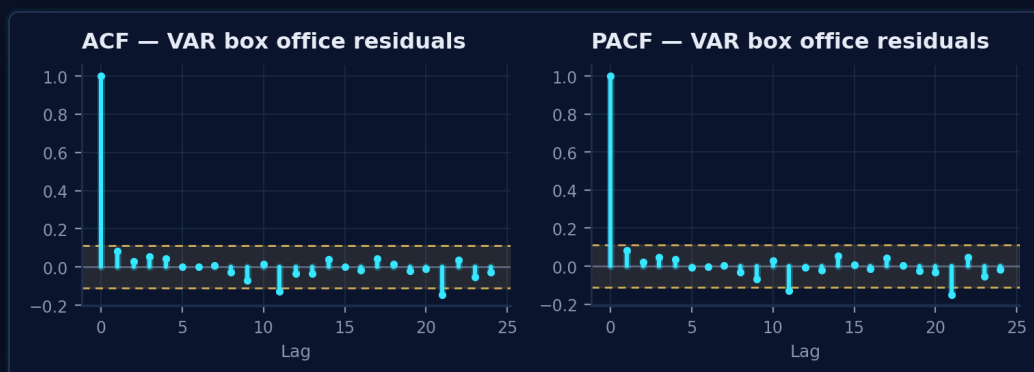


Figure G1–G2. Residual ACF/PACF for the VAR box office equation. Autocorrelation is largely contained within the confidence bands, though minor structure remains.

Limitations

Residual diagnostics show mild violations of homoskedasticity and no-autocorrelation, concentrated around the 2020 break. The models also omit industry-specific drivers—production cycles, streaming competition, individual tentpole releases—that materially shape revenue, bounding the explanatory power of macro variables alone. The natural extension adds a COVID regime dummy or break-adjusted specification and release-slate controls.

TAKEAWAY

The diagnostics are disclosed rather than buried: the framework is sound, the 2020 shock is the main source of strain, and the obvious next iteration is a regime-aware model with industry controls.

The film business sits at the intersection of macro forces and autonomy

What the analysis establishes, and why it matters commercially.

- **Consumer spending is the single most important macroeconomic determinant of box office revenue**, shaping both short-run fluctuations and the long-run equilibrium relationship.
- **Monetary policy matters only indirectly**, through its effect on discretionary income, and contributes little predictive power on its own.
- **The industry does not self-correct toward macro equilibrium**—the economy absorbs the adjustment—so theatrical revenue retains substantial autonomy driven by seasonality, content cycles, and external shocks.

For entertainment finance and strategy, the practical reading is direct: consumer-spending conditions belong in any serious theatrical revenue forecast or slate-financing model as a leading indicator, but they will never explain the majority of the variance. The industry's own release dynamics dominate—which is precisely why macro exposure cannot be fully hedged, and why greenlight and financing decisions must weight content-specific risk above the macro cycle.

BOTTOM LINE

Box office revenue responds to the macro economy without being governed by it. It is macro-sensitive in the short run, macro-independent in its correction, and industry-driven in its variance—a profile that rewards analysts who model the cycle and the slate together.

Methods: dynamic regression, three-lag VAR, Johansen cointegration, and VECM, estimated in R on 312 monthly observations (2000–2025). Figures reconstructed for presentation from the original model outputs; point estimates and qualitative conclusions unchanged. Prepared as an analytical work sample by Tanmay Srivastava.